Nate Annau

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Math and CS undergraduate interested in Computational Science, Computer Security, and Machine Learning

Education

University of California, Santa Barbara
Dual Bachelors of Science in Mathematics and Computer Science (3.87 GPA)
Sep 2022 — Jun 2026 Santa Barbara, CA

Sequoia High School
Jun 2020 — May 2022 Redwood City, CA
At Sequoia, I participated in Cañada Middle College, a full time Dual Enrollment program.
Thus, I attended high school classes simultaneously with Community College classes, so I could earn credit for university.
I received my diploma from Sequoia High School.

Mid-Peninsula High School
Sep 2018 — Jun 2020 Menlo Park, CA

Research Experience

University of California, Santa Barbara
Undergraduate Research Assistant
Jan 2026 — Present Santa Barbara, CA
Worked on a CUDA accelerated solver for Hamilton-Jacobi equations, as part of the Computational Applied Science Laboratory, under the supervision of Professor Frederic Gibou. My main work thus far is focused on generalizing the library to arbitrarily high dimensions. Met weekly with the research group to discuss progress.

University of California, Santa Barbara
Directed Reading Program (DRP) Participant
Jan 2025 — May 2025 Santa Barbara, CA
With fellow undergraduate Jesse Cobb, read *Geometric and Topological Aspects of Coxeter Groups and Buildings* by Anne Thomas and parts of *Reflection Groups and Coxeter Groups* by James Humphreys, under the supervision of PhD student Benedict Lee. Met weekly to discuss the readings, and presented the associated poster at the DRP showcase.

Work Experience

In the summer of 2026, I will be working on a cybersecurity research project in the United States Department of Energy, at Sandia National Laboratories.

Software Engineering Intern
Playlab AI
Jun 2025 — Sep 2025 Remote

- Developed custom benchmarking framework to statistically evaluate 10+ document parsing services across production documents, identifying optimal parsing strategies for RAG
- Expanded parsing and chunking capabilities significantly in the internal API, adding handling for common data formats
- Used heavy typing features in Python, focusing on maintainability and readability, and greatly expanded unit tests

Software Engineering Intern
HiveTop
Jun 2024 — Aug 2024 Berlin, Germany

- Developed key pages, such as a graph dashboard for device telemetry, of a frontend platform using `Vue` and `Tailwind` with `Primevue` components
- Created REST endpoints in the `Rust` backend to facilitate data transmission from embedded devices
- Architected major elements of our CI / CD setup, using Github actions, `ESLint`, and `mypy`

STEM Center Tutor
Cañada College
Jan 2022 — May 2022 Redwood City, CA

- Assisted students with homework and mastering key concepts in Calculus, Differential Equations, and Newtonian Mechanics
- Created high quality study materials and explanations to address common deficiencies in student understanding

Volunteering

Tutor
Healthy Cities Tutoring
Jan 2020 — May 2022 San Carlos, CA

Tutored middle school students in math.

Volunteer
Second Harvest Food Bank
Seasonal, 2019 — 2022 San Carlos, CA

Sorted and packaged food for low-income communities.

Projects and Competitions

Ergo

- In collaboration with my friend, developed a `Typst` package for notetaking and assignments for Math, CS, and Physics students
- The package is actively maintained, has 50+ stars on Github, and is being used for a PhD thesis at a French university
- Contains over a dozen custom environments, custom color schemes, and over one hundred macros for common symbols

Courses Graph

- Scraped the UCSB website with `Python`, queried the UCSB API, and synthesized the data as JSON
- Dynamically displayed course data using a constrained optimization library and `d3`, with `Quasar` components
- Presented working prototype in front of audience of peers and industry judges at the 2023 Coders SB Project Showcase, winning third place out of ~35 projects
- At SBHacks 2025, extended this general concept to a visualization of the dependency graph of Proof Wiki

Open Source Contributions

- Contributor to `weo-reader`, a Python client to read world economic data from the IMF as a `pandas` dataframe
- Authored just under one thousand Wikipedia contributions, and wrote several tools to help automate data collection
- Made corrections and wrote a chapter for a popular online summary textbook of early undergraduate math courses

CTFs (Cybersecurity Competitions)

- Won second place out of 300+ participants in the Geo OSINT challenge at DEF CON, the world's largest hacker conference
- As Team Captain, led a 10 person team to 12th place in iCTF 2026
- As the second highest scoring individual contributor on my CTF team, placed in the top 6% in the PwnMe CTF Qualifier
- Participated in 10+ CTFs in 2025, the second most of any team member. We ended 2025 as the 61st ranked team in the United States

Programming Languages and Tools

Python (numpy, pandas, Flask)	●●●●○
Git	●●●●○
Linux (Ubuntu, Arch, CentOS, Nix)	●●●●○
C++ and CUDA	●●●○○
TypeScript (Vue.js, Svelte)	●●●○○
Neovim	●●●○○
Rust	●●○○○

Languages

English – Native | **German** – Approx B1 | **Spanish** – A1